GCSE GEOGRAPHY

Paper 1: Living with the physical environment

Topic 2: The living world

Practice questions in the style of AQA GCSE Geography (8035). Every question is original; the layout, command words, mark allocations (including levels-based 6- and 9-mark questions with SPaG) and mark scheme follow the real exam format. Figures and data are illustrative and simplified for practice; case-study facts in the mark scheme are summarised, so check them against your own case-study notes.

Time allowed	About 120 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a black pen, a pencil, a ruler, a calculator and (for some questions) a protractor.
Instructions	• Use black ink or black ball-point pen. Use pencil only for drawing. • Answer **all** questions. • You must answer the questions in the spaces provided. • Do all rough work in this book. Cross through any work you do not want to be marked. • Questions that carry a **SPaG** note are also marked for spelling, punctuation, grammar and the use of specialist terminology.
Information	• The maximum mark for this paper is **116**. • The marks for questions are shown in brackets. • Questions that start with the command word *Explain* need developed points (a point plus a reason or consequence). • 6- and 9-mark questions are marked by **levels**: aim to give a detailed, balanced and well-linked answer, using place-specific detail.

Name: _____ Date: _____ Mark: _____ / 116

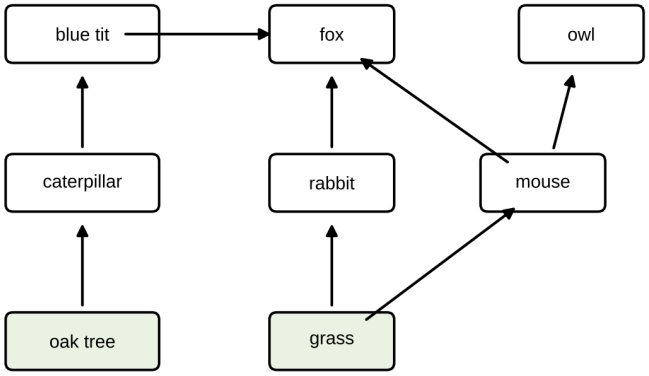
Section B: The living world

Answer all questions in this section.

0 1 This question is about ecosystems. **Figure 1** shows a food web in a UK woodland.

0 1 . 1 What is a **producer**? **[1 mark]**

Figure 1



*Tick **one** box.*

- [ ] an animal that eats only plants
- [ ] a plant that makes its own food by photosynthesis
- [ ] an organism that breaks down dead material
- [ ] an animal that eats other animals

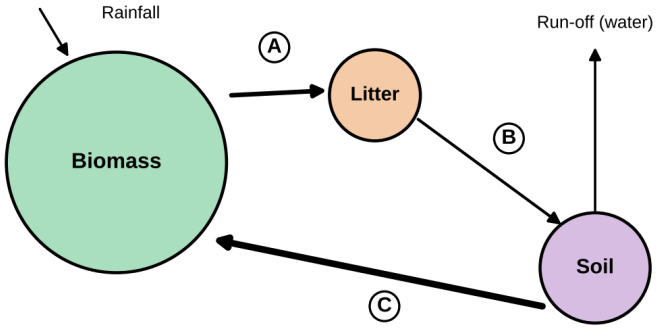
0 1 . 2 Name **one** producer and **one** secondary consumer shown in Figure 1. **[2 marks]**

0 1 . 3 Explain what might happen to the number of mice if the owls are removed from the woodland. **[3 marks]**

0 1 . 4 Describe the role of decomposers in an ecosystem. **[2 marks]**

0 1 . 5 **Figure 2** shows the nutrient cycle in a tropical rainforest. Name the stores labelled by the arrows **A** and **C**. **[2 marks]**

Figure 2



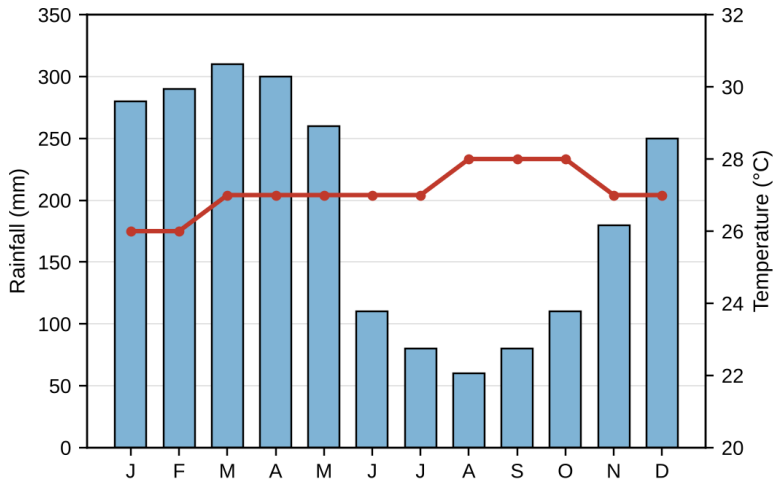
0 1 . 6 Explain why the biomass store is large and the litter store is small in a tropical rainforest. **[4 marks]**

0 1 . 7 Explain how a change in climate in one place could affect the plants and animals living there. **[4 marks]**

0 2 This question is about tropical rainforests. **Figure 3** shows the climate of a tropical rainforest location.

0 2 . 1 Describe the climate shown in Figure 3. **[4 marks]**

Figure 3

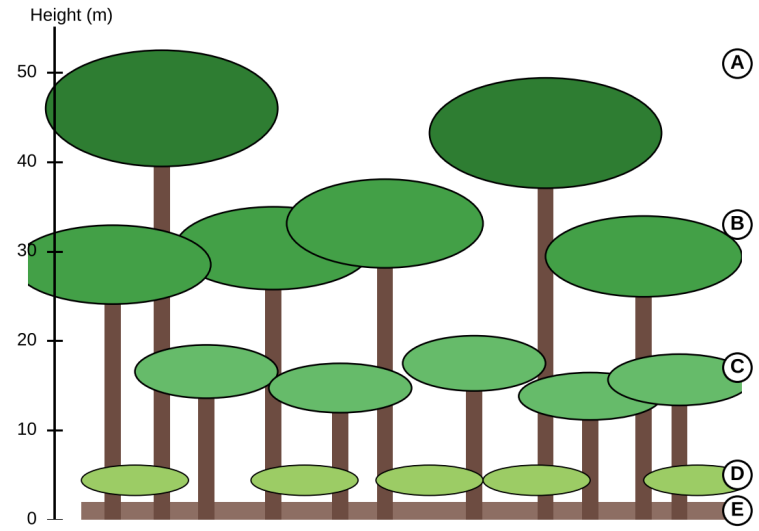


0 2 . 2 Calculate the annual rainfall total shown in Figure 3 (add all twelve months). **[2 marks]**

Answer = _____ mm

0 2 . 3 **Figure 4** shows the structure of a tropical rainforest. Name the layers **A** (emergent) and **C**. Describe one adaptation of plants in layer **A**. **[3 marks]**

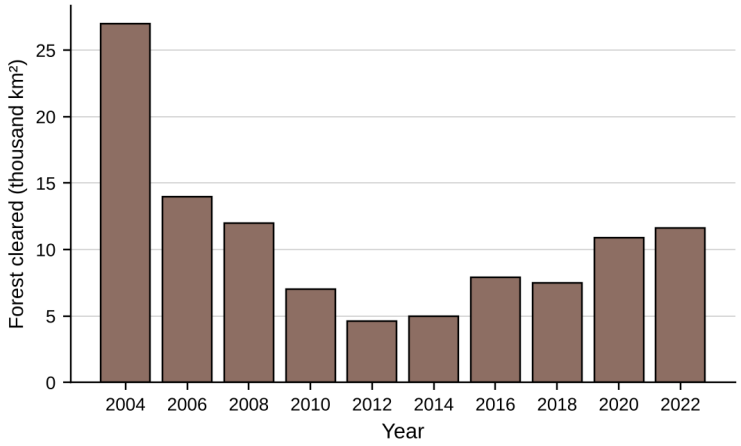
Figure 4



0 2 . 4 Explain how plants and animals are adapted to the tropical rainforest climate. **[4 marks]**

0 2 . 5 **Figure 5** shows how much forest was cleared in part of the Amazon. Describe the trend in deforestation between 2004 and 2022. **[3 marks]**

Figure 5



0 2 . 6 Explain **two** causes of deforestation in tropical rainforests. **[4 marks]**

0 2 . 7 Explain the impacts of deforestation on the rainforest environment and on the people who live there. **[4 marks]**

0 2 . 8 Explain how rainforests can be managed so that they are used sustainably. Refer to a named rainforest. **[6 marks]**

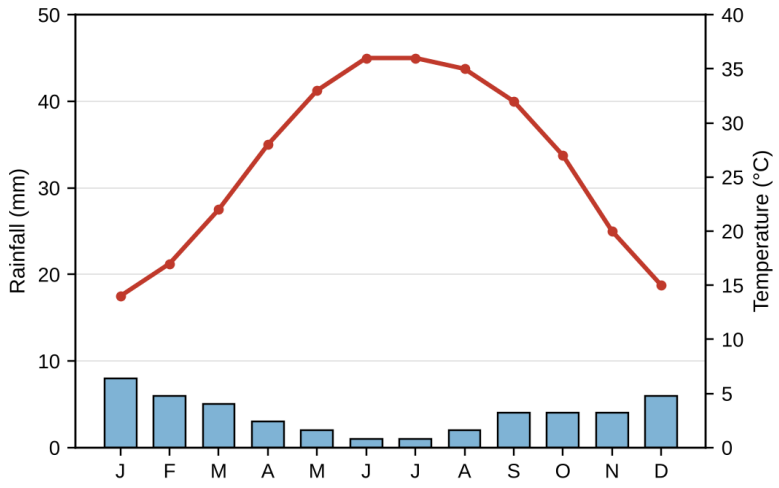
0 2 . 9 **'Tropical rainforests cannot be both used and protected.'** To what extent do you agree with this statement? Refer to a named rainforest. **[9 marks] + [3 marks] SPaG**

Spelling, punctuation, grammar and use of specialist terminology [3 marks]

0 3 This question is about hot deserts. **Figure 6** shows the climate of a hot desert location.

0 3 . 1 Give **two** characteristics of the climate shown in Figure 6. **[2 marks]**

Figure 6



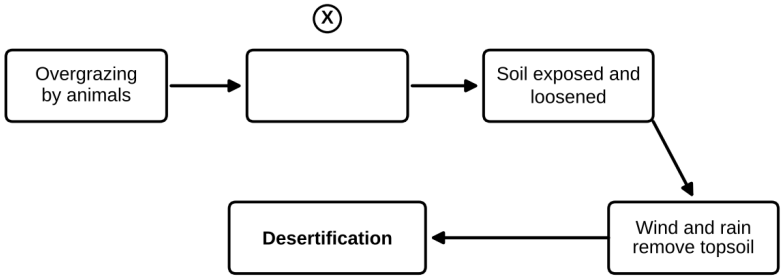
0 3 . 2 Explain how **two** plants or animals are adapted to living in a hot desert. **[4 marks]**

0 3 . 3 Describe **two** opportunities for economic development in a hot desert you have studied (e.g. Thar Desert). **[4 marks]**

0 3 . 4 Explain why development in hot deserts can be difficult. **[4 marks]**

0 3 . 5 Explain how overgrazing leads to desertification. Use Figure 7 to help you. **[4 marks]**

Figure 7



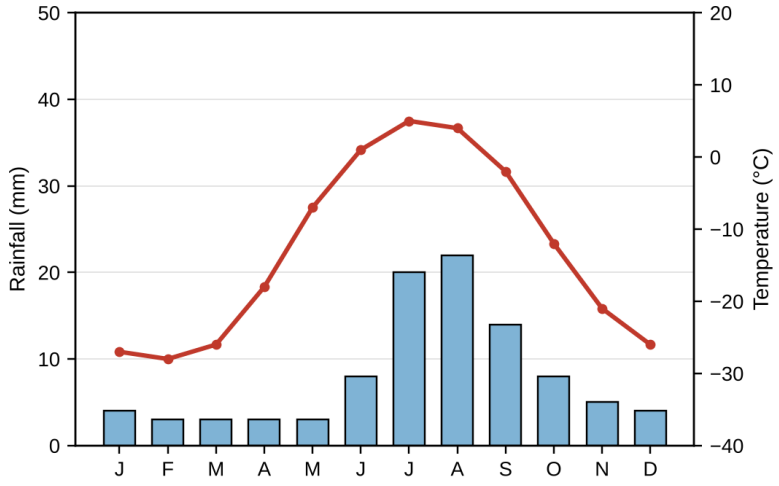
0 3 . 6 Describe **two** strategies that can be used to reduce the risk of desertification. **[4 marks]**

0 3 . 7 Explain the causes of desertification and the ways in which it can be managed. **[6 marks]**

0 4 This question is about cold environments. **Figure 8** shows the climate of a tundra location.

0 4 . 1 Describe the climate shown in Figure 8. **[3 marks]**

Figure 8



0 4 . 2 Explain how **two** animals or plants are adapted to life in a cold environment. **[4 marks]**

0 4 . 3 Give **two** reasons why cold environments are fragile. **[2 marks]**

0 4 . 4 Describe **two** opportunities for development in a cold environment you have studied (e.g. Alaska). **[4 marks]**

0 4 . 5 Explain how the Trans-Alaska Pipeline was designed to reduce the impact on the environment. **[3 marks]**

0 4 . 6 **'Cold environments should be protected from economic development.'** To what extent do you agree with this statement? Refer to a named cold environment. **[9 marks] + [3 marks] SPaG**

Spelling, punctuation, grammar and use of specialist terminology [3 marks]

END OF QUESTIONS

MARK SCHEME – Topic 2: The living world

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *geography*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	a living thing (plant) that makes its own food (using sunlight / photosynthesis)	Tick box 2 is correct.	1	AO1 3.1.2.1
01.2	producer: oak tree / grass ; secondary consumer: blue tit / fox / owl (eats other consumers) ;	Fox and owl are acceptable as secondary consumers (they eat primary consumers). Allow blue tit (eats caterpillars).	2	AO2 3.1.2.1
01.3	mice are no longer eaten by owls, so their numbers increase ; the mice eat more grass, so the amount of grass decreases ; numbers of foxes might increase / fox food supply changes (foxes eat more mice) ; later the mice may run out of food and numbers fall again ;	Any three linked points.	3	AO2 3.1.2.1
01.4	break down dead plants and animals / waste ; returning nutrients to the soil (so plants can use them again) ;		2	AO1 3.1.2.1
01.5	A: biomass to litter (leaf fall) ; C: soil to biomass (uptake by plants / roots) ;	Accept descriptions of the transfers (e.g. 'leaf fall', 'nutrient uptake'). Biomass = largest store in the rainforest.	2	AO2 3.1.2.1
01.6	hot, wet climate gives rapid plant growth, so a large biomass ; dead leaves / litter decompose very quickly ; because of the high temperature and moisture which help decomposers ; nutrients are taken up quickly by plants so soil store is small ;		4	AO2 3.1.2.1
01.7	climate controls vegetation growth (temperature and rainfall) ; if rainfall falls, plant growth falls / some plants die ; so herbivores have less food and numbers decline ; carnivores that eat the herbivores are then affected (food chain / interdependence) ;		4	AO2 3.1.2.1
02.1	temperature is high all year / 26–28 °C ; very small annual temperature range (about 2 °C) ; rainfall is high every month (60 mm or more) ; wetter months from December to May (peak about 310 mm in March) ; drier months July–September ; use of figures to support ;	Any four. Allow 'driest month August (60 mm)'.	4	AO3 3.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	280+290+310+300+260+110+80+60+80+110+180+250 ; 2 310 mm ;	Allow 2300–2320.	2	AO4 3.1.2.2
02.3	A: emergent layer ; C: understorey ; adaptation: very tall trees reach above the canopy for sunlight / trees have small waxy leaves to cope with winds and sun ;	A = emergent, B = canopy, C = understorey, D = shrub layer, E = forest floor.	3	AO1 3.1.2.2
02.4	plants have drip tips / waxy leaves so heavy rain runs off quickly ; buttress roots support tall trees in shallow soil ; epiphytes grow on branches to get light ; animals (e.g. monkeys / sloths) have adaptations such as camouflage or climbing limbs / prehensile tails ; plants grow tall to reach light (competition for light) ;	Each adaptation needs a reason (e.g. 'so that…').	4	AO2 3.1.2.2
02.5	large fall from about 27 000 km² (2004) to about 4 600 km² (2012) ; then increases again to about 11 600 km² in 2022 ; lowest in 2012 / highest in 2004 ; figures with units ;	Data are illustrative for practice.	3	AO3 3.1.2.2
02.6	cattle ranching: forest cleared for grazing land ; to meet beef demand ; soya / palm oil plantations ; cleared for commercial farming and export ; logging for hardwoods (mahogany) ; roads built to reach forest ; mineral extraction / mining and building of HEP dams ; subsistence farming / population growth ; settlement ;	Two causes, each with 1 mark for the cause and 1 for development.	4	AO1/AO2 3.1.2.2
02.7	loss of biodiversity / habitats for plants and animals ; soil erosion because roots no longer hold the soil / loss of nutrients when leaf litter disappears ; less CO_2 absorbed and more released when burned, adding to climate change ; indigenous people lose their homes / traditional way of life ; economic benefits: jobs and income from timber, farming, mining ;		4	AO1/AO2 3.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO2 3.1.2.2
	Indicative content (credit any other valid geography, and place-specific detail): • selective logging: only mature trees taken, replanted ; keeps the structure of the forest ; • replanting / reforestation and use of nurseries ; • ecotourism: visitors pay for guides / lodges, income for locals, little damage ; • international agreements (e.g. CITES, Amazon Fund) / debt reduction: reduces the need to clear land ; • protected areas / national parks and legislation (Brazil's Forest Code) ; • local community involvement / agro-forestry (growing crops under trees) ; • case-study detail: named place (e.g. Amazon, Brazil) and examples of schemes ;			
02.9	**Level 3 (7–9 marks)**: Detailed, accurate knowledge and understanding of the topic with a well-developed, balanced and logical argument. A clear, supported **evaluation/judgement** that is based on evidence (and specific place detail or the resources where asked). Any conclusion is fully justified. **Level 2 (4–6 marks)**: Some accurate knowledge and understanding. Some development of ideas with evidence; the judgement/evaluation is partial, unbalanced or not fully supported. **Level 1 (1–3 marks)**: Limited knowledge and understanding. Simple statements with little development; a judgement, if any, is unsupported. **0 marks**: No relevant content.		9 + 3 SPaG	AO1/AO2/AO3 3.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
	Indicative content (credit any other valid geography, and place-specific detail): • Agree: use (logging, farming, mining, roads) destroys habitat and biodiversity, and clearing reduces carbon storage ; profits often go to big companies / distant cities, not local people ; • Disagree: sustainable management shows both are possible – selective logging, ecotourism, agro-forestry, plantations on already-cleared land ; • Disagree: international agreements and debt reduction help countries earn income without destroying forest ; protected reserves ; • Balance: weak law enforcement and illegal clearance in the Amazon; short-term economic need vs long-term environment ; • Evaluation: it depends on scale, strict control and who benefits ; supported judgement with named example detail (e.g. Brazil's Amazon Fund) ;			
	SPaG (3 marks) – award separately: **High performance (3)**: Learners spell and punctuate with consistent accuracy and consistently use rules of grammar; they use a wide range of specialist terms adeptly and with precision. **Intermediate performance (2)**: Learners spell and punctuate with considerable accuracy and use rules of grammar with general control; they use a good range of specialist terms with facility. **Threshold performance (1)**: Learners spell and punctuate with reasonable accuracy and use rules of grammar with some control; they use a limited range of specialist terms appropriately. **0 marks**: No marks awarded: no response or the spelling, punctuation and grammar is too weak, or the terminology is not relevant.			
03.1	very hot summers (above 30 °C in June–September, peak about 36 °C) ; mild / warm winters (14–17 °C) ; large annual temperature range (about 22 °C) ; very low rainfall every month (less than 10 mm) ; annual rainfall total about 46 mm ;	Any two, with data for each.	2	AO3 3.1.2.3
03.2	cactus: thick, waxy stem stores water ; spines reduce water loss / protect ; long roots (shallow, wide) to absorb water quickly after rain ; camel: hump stores fat so it can go without food ; thick eyelashes keep out sand ; wide feet stop it sinking ; fennec fox: large ears lose heat ; nocturnal ; pale colour reflects sunlight ;	Two adaptations, each with 1 mark for the adaptation + 1 for the reason.	4	AO2 3.1.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	named desert (e.g. Thar Desert, Rajasthan, India) ; irrigation by canal (Indira Gandhi Canal) allows farming / crops such as wheat, cotton ; solar energy farms (e.g. Bhadla) use the sunshine ; mining (gypsum, limestone, building stone) ; tourism (e.g. Jaisalmer, camel safaris) ;	Two opportunities with detail: 1 mark each for the opportunity and 1 for description. Place detail needed for full marks.	4	AO1 3.1.2.3
03.4	extreme heat makes it hard to work / build / farm ; lack of water (little rainfall; water table low) ; poor soil / sand blown, few crops can grow ; remote, hard to reach – few roads / high cost of transport, power lines and pipes ; sandstorms damage equipment and roads ;		4	AO2 3.1.2.3
03.5	overgrazing: too many animals eat the vegetation ; box X: (vegetation / plant roots die and no longer hold the soil) ; soil is exposed and dries out, loosened and carried away by wind and rain ; soil loses its nutrients so land cannot support plants – becomes desert-like ;	What goes in X: 'vegetation removed / plants die'.	4	AO2 3.1.2.3
03.6	water management: drip irrigation / building dams / collecting rainwater ; soil management: stone lines / crop rotation / adding manure so soil stays fertile ; tree planting: root systems hold soil / windbreaks (e.g. Great Green Wall) ; appropriate technology: solar cookers reduce the need to collect wood ; controlled grazing: fewer animals, rotational grazing ;	Two strategies, each with 1 for the strategy + 1 for how it helps.	4	AO1 3.1.2.3
03.7	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO1/AO2 3.1.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
	Indicative content (credit any other valid geography, and place-specific detail): • causes: climate change / droughts ; population growth putting pressure on the land ; overgrazing ; over-cultivation ; removal of firewood / deforestation ; poor irrigation causing salinisation ; • so soil is exposed, eroded and loses fertility, vegetation cannot grow ; • management: water management (drip irrigation, dams) ; soil management (crop rotation, manure, stone lines) ; • tree planting (windbreaks / Great Green Wall) ; appropriate technology (solar cookers) ; • evaluation: strategies need cooperation, cost / finance and local involvement ; low-cost local measures are often more sustainable ;			
04.1	very cold winters / below −25 °C (October–April / −28 °C in February) ; short cool summer (only June–September above 0 °C; peak 5 °C in July) ; low rainfall / precipitation (often 3–4 mm in winter), small peak in July–August (22 mm) ; large annual temperature range / use of figures ;	Any three, with data.	3	AO3 3.1.2.4
04.2	polar bear: thick fur and layer of fat (blubber) for insulation ; white fur for camouflage ; arctic fox: small ears / thick fur reduce heat loss ; white coat camouflage in winter ; tundra plants: low-growing / grow in clumps to avoid wind ; small leaves reduce water loss ; short growing season so grow and seed quickly ;	Two adaptations, 1 mark for the adaptation + 1 for the reason.	4	AO2 3.1.2.4
04.3	slow plant growth because of the short growing season, so damage takes decades to recover ; food chains are simple – loss of one species affects many others ; melting permafrost damages soil and releases carbon ; pollution (oil spills) lasts for a long time in the cold ;		2	AO1 3.1.2.4
04.4	named place e.g. Alaska ; oil and gas (Prudhoe Bay) – jobs and income ; mineral extraction (gold, zinc) ; tourism (wilderness, cruises, wildlife, Denali National Park) ; fishing (salmon) ;	1 mark for the opportunity + 1 for detail, for two opportunities.	4	AO1 3.1.2.4
04.5	raised on stilts / supports (above ground) ; so heat from the oil does not melt the permafrost and cause the pipe to sink / break ; built with a zig-zag shape, to allow expansion and contraction / gaps for animals such as caribou to cross ;	Accept 'built with insulation' for 1 mark.	3	AO2 3.1.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	**Level 3 (7–9 marks)**: Detailed, accurate knowledge and understanding of the topic with a well-developed, balanced and logical argument. A clear, supported **evaluation/judgement** that is based on evidence (and specific place detail or the resources where asked). Any conclusion is fully justified. **Level 2 (4–6 marks)**: Some accurate knowledge and understanding. Some development of ideas with evidence; the judgement/evaluation is partial, unbalanced or not fully supported. **Level 1 (1–3 marks)**: Limited knowledge and understanding. Simple statements with little development; a judgement, if any, is unsupported. **0 marks**: No relevant content.		9 + 3 SPaG	AO1/AO2/AO3 3.1.2.4
	Indicative content (credit any other valid geography, and place-specific detail): • Agree: fragile ecosystems with slow recovery ; permafrost melt releases greenhouse gases ; oil spills and pollution harm wildlife ; remote wilderness would be lost ; • Agree: indigenous people's traditional lifestyle (e.g. Inuit) threatened ; global importance of Arctic / Antarctic (e.g. Antarctic Treaty protects Antarctica) ; • Disagree/balance: development brings jobs, income and energy – oil revenues support Alaskan services ; technology (pipeline on stilts, strict rules) reduces damage ; • Balance: tourism can fund conservation but brings waste / pollution ; wilderness areas and national parks allow limited, controlled use ; • Judgement: protected areas + controlled development, with evidence from Alaska / Antarctica / Svalbard ;			
	SPaG (3 marks) – award separately: **High performance (3)**: Learners spell and punctuate with consistent accuracy and consistently use rules of grammar; they use a wide range of specialist terms adeptly and with precision. **Intermediate performance (2)**: Learners spell and punctuate with considerable accuracy and use rules of grammar with general control; they use a good range of specialist terms with facility. **Threshold performance (1)**: Learners spell and punctuate with reasonable accuracy and use rules of grammar with some control; they use a limited range of specialist terms appropriately. **0 marks**: No marks awarded: no response or the spelling, punctuation and grammar is too weak, or the terminology is not relevant.			

Where marks are lost in Topic 2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Climate graphs:** give figures with units for *both* temperature and rainfall, name the months, and give a range or total where it helps. Only then add description words like 'hot and wet'.
- ☐ **Adaptations:** name the plant/animal, the adaptation, and *why it helps* (e.g. 'drip tips so water runs off quickly').
- ☐ **Nutrient cycle:** the size of the circle shows the store size; the thickness of the arrow shows the speed of the transfer. In the rainforest the biomass store is biggest.
- ☐ **Case-study detail:** name the place (Amazon / Thar / Alaska) and give at least two pieces of place-specific detail in 6- and 9-mark answers.
- ☐ **Causes vs impacts:** do not mix them. Plan 'causes → impacts → management' in that order.
- ☐ **Management answers:** say how the strategy works *and* whether it is sustainable (can it continue without damaging the environment for the future?).
- ☐ **Evaluation:** '9 marks' needs both sides plus a conclusion – 'On balance… because…'.
- ☐ **Desertification:** be clear that it is caused by both natural (climate) and human (overgrazing) factors.